

Phase 1: Setup & Data

- Set up GitHub repository
- Load the customer support tickets dataset
- Explore data distributions
- Basic preprocessing (lowercasing, removing stopwords, punctuation, mistakes and more)
- Write EDA summary with plots (word frequency, ticket volume over time, etc.)

Phase 2: Clustering Analysis

- Implement k-means clustering on TF-IDF vectors
- Implement k-means clustering on embeddings
- Implement HAC (Hierarchical Agglomerative Clustering) as comparison
- Investigate stability across different values of k
- Evaluate cluster coherence and interpretability
- Document findings — do clusters make intuitive sense?

Phase 3: Define Analytic Axes

- Choosing our 2 analytic axes
- Write clear hypotheses for each approach before running experiments
- Document reasoning for chosen axes in the report draft

Phase 4: Analytics Pipeline

- Implement Axis 1 approach A (TF-IDF representation)
- Implement Axis 1 approach B (embeddings)
- Implement Axis 1 approach C if needed (topics/phrases)
- Implement Axis 2 approach A (sentiment analysis labelling)
- Implement Axis 2 approach B (topic modelling for themes)
- Implement Axis 2 approach C if needed
- All the previous brackets are examples
- Identify and rank main issue types or themes by importance
- Explore temporal trends if applicable

Phase 5: Evaluation

- Measure consistency and robustness of findings across methods
- Measure coherence and interpretability of issue categories
- Check alignment with initial clustering results
- Check alignment with intuitive/expected patterns
- Error analysis — where do methods disagree or fail?

Phase 6: Report Writing

- Abstract (0.25 pages)
- Introduction with data examples and EDA plots (1.5 pages)
- Methods section — baseline, axes, hypotheses, improvements (3.5 pages)
- Experimental evaluation — metrics, setup, results tables/plots, error analysis, discussion (3.5 pages)
- Conclusions (1 page)
- Proofread and format using NeurIPS or ACL Overleaf template
- Add GitHub link to report
- Submit group report to Turnitin by 13:00, May 1th

Phase 7: Individual Reflections (each person separately)

- Write contributions summary with GitHub activity screenshot (max 150 words)

- Write technical understanding section — one challenge you worked on (max 300 words)
- Write teamwork section — roles and coordination tools (max 200 words)
- Submit individual reflection to Turnitin by 13:00, May 1th

Ongoing throughout:

- Make sure to meet 1-2 times a week minimum
- Attend all the supervisor meetings and maybe we can also get some feedback of him
- Keep GitHub commits regular so activity is visible for the reflection